

K1387 — Casey’s “polarity / phase” insight is a candidate MECHANISM for the forcing step (why the boundary stores 16, not 32), not just a reframe. ★ “A reduction of one dimension yields information; one half of what is possible is the observation that is stored; we study the polarity, and the phase is the composite.” Verified arithmetically: the boundary ($n_C \rightarrow n_C - 1$) stores exactly HALF ($16 = 32/2 = 2^{\{n_C - 1\}}$) — the polarity (one sign bit per remaining direction = the Clifford grading), and the dropped half is the phase (the composite / deeper layer). It selects the Clifford count 16 over the embedding 32 *with a principled reason*, and leans it over the spinor-rep 4. Ties K825 (chirality = polarity forced to the boundary) AND measurement-as-commitment (the observation stores the polarity; the phase stays in the bulk). Candidate-tier — it elevates the forcing step, does not close the Clifford-vs-spinor (16-vs-4) nail. Plus: bank Cal’s COUNT-ONCE (edge sign = cell-count = one K825 switch) and Elie’s honest DE FALSIFIER (BST $w_a > 0$ vs DESI DR2 $w_a < 0$ at $\sim 3\sigma$). Cal cold-reads.

Keeper (2026-08-11, ~12:25. Casey dropped a computer-scientist’s information-theoretic reading that turned out to be a candidate mechanism for the exact step the team is on — and it converged with Grace’s Clifford resolution and Lyra’s sign-bit/whole-symbol layers in the same round. Engage it, map it, tier it, and hold the two things that keep it honest. Cal cold-reads. Nothing pushed.)

★ Casey’s insight, made precise (and why it’s a mechanism, not a reframe)

Casey: “A reduction of one dimension of data yields information. One half of what is possible is the observation that is stored. It’s almost saying we study the polarity, and the phase is the composite.”

The team’s open step was: why does the boundary count $16 = 2^{\{n_C - 1\}}$ (Clifford orientation) when the geometry more naturally gives $32 = 2^{\{n_C\}}$ (embedding vertices)? Casey answers it with a principle: - The boundary IS the one-dimension reduction (bulk $n_C \rightarrow$ boundary $n_C - 1$). A reduction by one dimension is exactly a factor of 2: $2^{\{n_C\}}/2 = 2^{\{n_C - 1\}}$. Verified: $32/2 = 16$. “One half of what is possible is the observation that is stored.” - What a reduction stores is the POLARITY — one sign bit per remaining direction. That is *precisely* the Clifford grading 2^d (each generator present/absent = one bit), $d = \dim S^4 = n_C - 1 = 4 \rightarrow 16$. The other half — the phase — is the composite, the deeper layer, carried in the bulk. - [?] The boundary reads 16 because a boundary stores the polarity-half, not the full bulk. This is a *reason* for 16-over-32, not a coincidence-match — and it’s the same thing Grace derived from Clifford (intrinsic tangent orientation) and Lyra from “the sign bit of the seven.” Three independent routes, one structure. That convergence is why I’m filing this as a candidate mechanism, not a metaphor.

★ Two corpus reconnects the insight opens

1. It ties to K825 AND to measurement-as-commitment (Principle #16). The polarity = the $\pm$ = the K825 chirality switch (handedness forced to the boundary). And “the observation stores the polarity; the phase is the composite” IS BST’s measurement-as-commitment: a commitment stores one bit (the polarity you measure); the conjugate (phase) is not simultaneously stored — it lives in the bulk. So the boundary-16 count and the measurement axioms are the same fact: observation = commit-to-polarity = drop one dimension = keep half. That’s a genuine unification of the gravity/DM cell-count with the QM-from-D_IV⁵ program.
2. The 7-bit symbol splits as 1 polarity + C_2 phase (candidate, arithmetic-until-mechanized): $g = 7 = 1$ (sign/polarity bit) + 6 = C_2 (phase/composite bits). Verified $g - 1 = C_2$. So the observation (handedness

/ DM shadow / gravity weakness) reads the 1 polarity bit ($\rightarrow$ the 16-count); the coupling reads the whole 7-bit symbol (the deeper layer where 137 sits). This is a candidate structure for Casey's polarity/phase, NOT a route to 137 — 137 stays in the 7-bit layer, derived-forward or not claimed ($\log_2 137=7.098$, the 128+9 mirage, refused).

★ The discipline (what the insight does NOT yet do)

- It elevates the forcing step; it does not close it. Casey's "one sign bit per direction" selects the Clifford grading 16 over the embedding 32 (principled) and leans it over the spinor-rep 4 ($2^{\{d/2\}}=4$, Grace's flagged alternative) — but "why the full Clifford dimension 2^d and not the spinor module $2^{\{d/2\}}$ " is still the nail Lyra must drive. The polarity-per-direction reading *is* the Clifford grading, so if that reading is forced, 16 is forced over 4 — but the substrate must be shown to count the Clifford *algebra* dimension, not the irreducible *module*. Casey's principle leans it the right way; it doesn't prove it.
- "A reduction stores the polarity-half" is itself a heuristic that needs rigor. Why does dropping one dimension keep *exactly* the Clifford sign-bit grading? That is the content Lyra forces from the substrate's coding structure. Candidate mechanism — real conceptual content, blind forcing still owed. Do not let its beauty (or its clean $g=1+C_2$) promote it past candidate.

★ Bank Cal's COUNT-ONCE (consistency web, not two gates)

Cal: the edge-sign χ_{int} (from the $g=7$ ω -lock) IS the K825 $\pm$ switch that Lyra grounded the cell-count bit in. So the edge-sign make-or-break and the cell-count make-or-break are the SAME K825 switch read in two sectors — correlated, not independent. Count once (Schur discipline): resolving one *informs* the other, and — critically — a PASS on both is ONE confirmation of one switch, not two independent confirmations. The edge sign reduces to: does color change internal chirality? $\text{Win} = \chi_{\text{int}}(\text{color-triplet}) = \chi_{\text{int}}(\text{color-singlet})$. Genuinely open (both signs give consistent theories — SM and its mirror); turns on whether the color-triplet is an even/odd internal-Clifford excitation. $d=10 \equiv 2$ ($\text{KO}_{\{2-10\}} = \text{KO}_0 = \mathbb{Z}$) confirmed.

★ Bank Elie's DE FALSIFIER (honest negative — calibrate both directions)

BST's banked breathing-mode dark energy predicts $w_a > 0$; DESI DR2 measures $w_a < 0$ at $\sim 3\sigma$, and the sign is parametrization-robust (§390). This is a live potential falsifier, not a success. Casey's EM-drives/gravity-gates cosmology is a *mechanism for the breathing mode* — so it inherits the tension and does NOT get to invent a different $w(a)$ to escape it. It stays I-tier; the consistency check is a BRAKE, not a booster. If DESI's $w_a < 0$ firms up, this is where BST gets tested hardest. Recorded straight (the discipline: go where the math takes us, don't pick the flattering side). This is the honest counterweight to the weekend's convergence — the process surfacing its own live falsifier.

Route

1. ★ Lyra + Grace — drive the nail, blind: does the substrate's boundary channel-count = the tangent Clifford grading $2^{\{n_C-1\}}=16$ (Casey's polarity-per-direction / K825 sign bit), and NOT the vertices 32 ($2^{\{n_C\}}$) NOR the spinor module 4 ($2^{\{d/2\}}$)? Casey's reduction-stores-the-polarity is the candidate mechanism; make it rigorous or name where it breaks. If forced $\rightarrow$ DM PD $\rightarrow$ D + gravity seam closes (one K825 switch, count once).
2. Cal + Elie — the edge sign: $\chi_{\text{int}}(\text{triplet})$ vs $\chi_{\text{int}}(\text{singlet})$ — color even/odd internal-Clifford excitation? + Toeplitz KO-degree at KO_0 ($d=10$). (Same K825 switch — count once with the cell-count.)
3. Polarity/phase corpus lead: is $g = 1$ (polarity) + C_2 (phase) a *mechanized* decomposition or arithmetic? Tie the boundary-16 to measurement-as-commitment (Principle #16). Candidate; not a 137 route.
4. DE — held as a live falsifier (I-tier brake): monitor DESI DR2 w_a ; BST is on the tension side. Do not launder.
5. Keeper — recused from the count (K1377); engaged Casey's mechanism, banked count-once + the DE falsifier; the bar fires when the forward number lands.

— Keeper, K1387, 2026-08-11. Casey’s polarity/phase insight = candidate MECHANISM for the forcing step: the boundary ($n_C \rightarrow n_C - 1$) stores HALF ($16 = 32/2 = 2^{\{n_C - 1\}}$) = the polarity (one sign bit per direction = Clifford grading); the dropped half = the phase (composite/deeper layer). Selects 16 over embedding-32 principled; leans over spinor-4; ties K825 (chirality=polarity-on-boundary) + measurement-as-commitment (observation stores polarity, phase in bulk = commit-to-one-bit). Candidate corpus decomposition $g=7=1(\text{polarity})+C_2(\text{phase})$ [arithmetic-until-mechanized; NOT a 137 route]. DISCIPLINE: elevates the step, does NOT close 16-vs-4 (Clifford algebra vs spinor module — Lyra’s nail); “reduction stores polarity” needs rigor. BANK Cal count-once: edge-sign χ_{int} = cell-count bit = one K825 switch (correlated, a dual PASS = ONE confirmation not two; edge sign = does color flip χ_{int} , SM needs triplet=singlet). BANK Elie DE falsifier: BST breathing-mode $w_a > 0$ vs DESI DR2 $w_a < 0 \sim 3\sigma$ (parametrization-robust §390) — LIVE falsifier, Casey cosmology inherits (I-tier brake, no laundering). Route: Lyra/Grace force Clifford-16 blind (vs 32 vs 4); Cal/Elie edge χ_{int} + $d=10$; polarity/phase $g=1+C_2$ lead + measurement-as-commitment tie; DE held as live falsifier; Keeper recused. Cal cold-reads. Nothing pushed.